**Astronomy 302: Introduction to Observational Astronomy**
Fall 2026 Semester

**Lectures:** MWF 10:00 -10:50am **Location:** SO 204

## Course Description

Students will learn how to develop and execute an observing program with optical/infrared telescopes. Experience with computers and some python is required. We will cover how various types of telescopes and their instruments work to take data across the electromagnetic spectrum in various formats. Students will learn about measurement error and error propagation along with its application in interpreting astronomical observations. Weather permitting, students will execute their own observing program in small groups using the 61" Kuiper Telescope. Students will use software and create python-based code to analyze data from their observing run or archival data. Students will write up their results and present their projects to the class in small groups.

## Instructors

**Prof. Brittany Miles (she/her)**
Office: SO 486
Email – bemiles@arizona.edu
Office Hours – by appointment

**Dr. Peter Milne (he/him)**
Office: SO N508
Email - pmilne@arizona.edu
Office Hours – TBD

## Teaching Assistants

**Jialin Li (she/her)**
Office: SO N412
Email – jialinli@arizona.edu
Office Hours – TBD

**Evan Mayer (he/him)**
Office: SO 341
Email - evanmayer@arizona.edu
Office Hours – TBD

## Observers

**Zachary Haarz (no preference)**
Email - zsh2027@arizona.edu

## Expected Outcomes and Goals

At the end of the class students will be able to:

- **Understand measurement precision, accuracy, and uncertainty.**
- **Locate astronomical targets on the celestial sphere. Understand how certain astronomical objects move over time.**
- **Describe how telescopes collect light and instruments record imaging or spectroscopic data.**
- **Search for various types of astronomical data:**
    - Coordinate information of targets
    - Photometric and/or telluric calibration stars
    - Archival detector/receiver images from different observatories
- **Create a plan to execute an observational or archival astronomical program.**
- **Visualize data from astronomical instruments using software and python code.**
- **Reduce astronomical data using Jupyter notebooks or python scripts.**

These learning outcomes will be met through the attendance of lectures, writing assignments, Python notebook problem sets, an independent, data reduction and analysis research project, and in-class discussions.

## Grades

Your final course grade will be the result of assessments of the **regular workbook/written assignments (60%) and the final project plus presentation to the class (40%).** The workbook/written assignments are to be done independently, but discussions with other students, the TA, or instructor are allowed. If you have a true emergency and cannot attend the final presentation of your project, please contact me immediately with documentation. Late work will not be accepted unless agreed upon with the professor beforehand. Ask for an extension at least a day before the deadline. If we do not receive assignments on time, we cannot provide sufficient feedback or adjustments to meet your needs as students. Workbook grades will be based on demonstrated effort, completion, and correctness of the results, and on clarity/accessibility of presentation, including documentation of any code. The final research project will be done in teams of ~4, but each student will turn in their own project report.

**A:** 90-100%, **B:** 80-89%, **C:** 70-79%, **D:** 50-69%, **E:** 0-49%

## Participation

Your engagement, quality of contributions, and interactions with the instructors and with your peers will contribute to your final grade. Quality and frequency of your contributions will be evaluated throughout the course, rewarding specifically **growth** and **improvement**.

**Class Discord Link - https://discord.gg/UQAxx6xzC**

## Final Project

A significant fraction of the course grade will be your independent observing program. This will be on a scientific question of your choice. The topic will be submitted to the professor for approval and/or revision. We will discuss opportunities during the class, but you are also welcome to choose an entirely different topic. It can either utilize archival data or data obtained at our telescopes. The project grade will consist of:

**80%:** A 5-page minimum science write-up describing the motivation (i.e., why it is interesting and what you expected to learn from it), a description of the method/equations that you used, comparison with relevant observations or theoretical expectations, your conclusions, and a discussion of how this program could be improved in the future. This write-up should include embedded references. Additional pages (unlimited) to provide any specialized code or other supporting materials.

**15%**: The team will give a final 15-minute in-class presentation covering the same topics (intro, methods, comparison, future).

**5%**: Early in the semester, the team will give a 5 min in-class presentation of the proposed project.

Relevant Reading: How to Plan your Astronomy Research paper in Ten Steps

## Required Materials

For this class you will need daily access to the following hardware: computer (laptop since you will bring it to class); regular access to reliable internet signal; ability to download and run the following software: web browser, zoom, and python. You will want to run the notebooks on your own machine, in which case you'll need Python and Jupyter Notebook, but this is not required for the final project (and the TA and I will not necessarily be able to support your private Python installation).

## Textbooks

Textbooks are not required, but I will be using information from the following books

"**An Introduction to Error Analysis: The Study of Uncertainties in Physical Measurements**" – John R. Taylor 2nd Edition

"**Observational Astronomy**" – D. Scott Birney, Guillermo Gonzalez, David and Oesper 2nd Edition

## Generative AI Use

In this course any and all uses of generative artificial intelligence (AI)/large language model tools such as ChatGPT, Claude, Dall-e, Google Bard, Microsoft Bing, etc. will be considered a violation of the Code of Academic Integrity, specifically the prohibition against submitting work that is not your own. This applies to all assessments in the course, including case studies, written assignments, discussions, quizzes, exams, and problem sets. This course policy is driven by the learning goals and desired learning outcomes for the course described above.

The following actions are **prohibited**:

- entering all or any part of an assignment statement or test questions as part of a prompt to a large language model AI tool.
- incorporating any part of an AI-written response in an assignment.
- using AI to summarize or contextualize reading assignments or source materials and
- submitting your own work for this class to a large language model AI tool for iteration or improvement.

## Tentative Schedule

| Date | Week | Topic | Observing and Notes | Assignment Due |
|---|---|---|---|---|
| Aug 24 2026 | 1 | Introduction and Syllabus Review | | |
| Aug 26 2026 | 1 | Measurement and Measurement Error | | |
| Aug 28 2026 | 1 | **Python Intro: Manipulating Images, Probability Distributions** | | **Read Procedures for 61" and 90" Telescopes/Project group and Availability Submission** |
| Aug 31 2026 | 2 | Uncertainties and Statistical Distributions | | |
| Sep 2 2026 | 2 | Coordinates and the Celestial Sphere | | |
| Sep 4 2026 | 2 | **Computer Work Day - Target Visibility Tools, Astroplan** | | |
| Sep 7 2026 | 3 | **Holiday (NO CLASS)** | | |
| Sep 9 2026 | 3 | Estimating SNR and Time Required for Observing | | |
| Sep 11 2026 | 3 | Detectors 1: Semiconductors and Bolometers | | **Observing Proposal Due** |
| Sep 14 2026 | 4 | Detectors 2: Read Out Electronics and Gain | | |
| Sep 16 2026 | 4 | Detector + Instrument Calibrations: Flat Fielding + Bias Frames | | |
| Sep 18 2026 | 4 | **Computer Work Day - Neptune and Uranus Lab** | | **Submit Filter + Flat Field Lab to D2L** |
| Sep 21 2026 | 5 | Optical Telescopes | | |

| Date | Week | Topic | Observing and Notes | Assignment Due |
|---|---|---|---|---|
| Sep 23 2026 | 5 | Photometry | | |
| Sep 25 2026 | 5 | **Computer Work Day - Photometry Lab** | | **Proposal Feedback to Peers** |
| Sep 28 2026 | 6 | Spectroscopy + Spectrographs | | |
| Sep 30 2026 | 6 | Spectrographs: Resolution and Dispersive Elements | | |
| Oct 2 2026 | 6 | **Computer Work Day - Photometry Lab** | **Observing** | **Submit Photometry Lab on D2L** |
| Oct 5 2026 | 7 | Spectrographs: Properties and Calibrations | | |
| Oct 7 2026 | 7 | Integral Field Spectrographs | | Dr. Miles may be OUT |
| Oct 9 2026 | 7 | **Computer Work Day - Spectroscopy Lab/Using Nimoy** | **Observing** | |
| Oct 12 2026 | 8 | Earth's Atmosphere and Adaptive Optics | | |
| Oct 14 2026 | 8 | buffer | | |
| Oct 16 2026 | 8 | **Computer Work Day - Spectroscopy Lab** | **Observing** | **Submit Spectroscopy Lab on D2L** |
| Oct 19 2026 | 9 | Considerations for Observatory Location and Hardware | | |
| Oct 21 2026 | 9 | Least Squares Fitting | | |
| Oct 23 2026 | 9 | **Fitting Data to Models** | | |
| Oct 26 2026 | 10 | Infrared Astronomy | | |
| Oct 28 2026 | 10 | Infrared Astronomy cont | | |
| Oct 30 2026 | 10 | **Time-Series or High Contrast Imaging Lab** | **Observing** | **Submit Data Fitting Lab on D2L** |
| Nov 2 2026 | 11 | Scientific Writing | | |
| Nov 4 2026 | 11 | Mock Journal Club | Day after midterm elections | |
| Nov 6 2026 | 11 | **Computer Work Day - Overleaf and Paper Reflection** | | |
| Nov 9 2026 | 12 | buffer | | |
| Nov 11 2026 | 12 | **Holiday (NO CLASS)** | | |
| Nov 13 2026 | 12 | | | **Submit Paper Reflection** |
| Nov 16 2026 | 13 | | | |
| Nov 18 2026 | 13 | | | |
| Nov 20 2026 | 13 | **Computer Work Day - Project Work** | | **Submit Initial Data Reduction Plot with Axis Labels and Units** |
| Nov 23 2026 | 14 | Radio Astronomy and Interferometry | | |
| Nov 25 2026 | 14 | **Computer Work Day - Project Work** | | |
| Nov 27 2026 | 14 | **Holiday (NO CLASS)** | | |
| Nov 30 2026 | 15 | High Energy Astronomy (Gamma, X-ray, UV) | | **Submit 10 minute Presentations** |
| Dec 2 2026 | 15 | **Computer Work Day - Project Work** | | |

| Date | Week | Topic | Observing and Notes | Assignment Due |
|---|---|---|---|---|
| Dec 4 2026 | 15 | **Computer Work Day - Project Work** | | |
| Dec 7 2026 | 16 | Project Presentations | | |
| Dec 9 2026 | 16 | Project Presentations (Last Day of Class) | | |
| Dec 11 2026 | 16 | **Submit Final Reports** | | **Submit Write Ups** |

### Classroom Behavior

To foster a positive learning environment, students and instructors have a shared responsibility. We want a safe, welcoming, and inclusive environment where all of us feel comfortable with each other and where we can challenge ourselves to succeed. To that end, our focus is on the tasks at hand and not on extraneous activities (e.g., texting, chatting, web surfing, etc.). Students are asked to refrain from disruptive conversations with people sitting around them during lecture. Students observed engaging in disruptive activity will be asked to cease this behavior. Those who continue to disrupt the class will be asked to leave lecture or discussion and may be reported to the Dean of Students.

The Arizona Board of Regents' Student Code of Conduct, ABOR Policy 5-308, prohibits threats of physical harm to any member of the University community, including to one's self. See: http://policy.web.arizona.edu/threatening-behavior-students

Any social media groups generated using university-sponsored tools (e.g., D2L class lists, discord servers) should be treated as an extension of the classroom. This means that they will need to abide by the University's student code of conduct policies, and be respectful, inclusive environments.

### Nondiscrimination and Anti-harassment Policy

The University of Arizona is committed to creating and maintaining an environment free of discrimination. In support of this commitment, the University prohibits discrimination, including harassment and retaliation, based on a protected classification, including race, color, religion, sex, national origin, age, disability, veteran status, sexual orientation, gender identity, or genetic information. For more information, including how to report a concern, please see http://policy.arizona.edu/human-resources/nondiscrimination-and-anti-harassment-policy

### Accessibility and accommodations

At the University of Arizona, we strive to make learning experiences as accessible as possible. If you anticipate or experience barriers based on disability or pregnancy, please contact the Disability Resource Center (520-621-3268, https://drc.arizona.edu/) to establish reasonable accommodations.

### Class recordings

To facilitate everyone's participation, audio and/or video recording of in class activities is not allowed unless explicitly agreed upon by all present students *and* instructors. In no case sharing on public platforms of the recordings will be allowed.

### Preferred Name and Pronouns

This course affirms people of all gender expressions and gender identities. If you prefer to be called a different name than what is on the class roster, please let me know. Feel free to correct instructors on your preferred gender pronoun.

### Attendance Policy

All holidays or special events observed by organized religions will be honored for those students who show affiliation with that particular religion. Absences pre-approved by the UA Dean of Students (or Dean's designee) will be honored. It is important to attend all classes, as what is discussed in class is pertinent to adequate performance on assignments and exams. If you must be absent, it is your responsibility to obtain and review the information you missed.

**Academic Integrity**
Integrity is expected of every student in all academic work. The guiding principle of academic integrity is that a student's submitted work must be the student's own. Students are encouraged to share intellectual views and discuss freely the principles and applications of course materials. However, **graded work/exercises must be the product of your own effort unless otherwise instructed**. Students are expected to adhere to the UA Code of Academic Integrity as described in the UA General Catalog. See: https://deanofstudents.arizona.edu/student-rights-responsibilities/academic-integrity

Misappropriation of exams before or after they are given will be considered academic misconduct. Misconduct of any kind will be prosecuted and may result in any or all of the following:

- Reduction of grade
- Failing grade
- Referral to the Dean of Students for consideration of additional penalty, i.e. notation on a student's transcript re. academic integrity violation, etc. http://deanofstudents.arizona.edu/policies-and-codes/code-academic-integrity

**Additional resources for students**
UA Academic policies and procedures are available at http://catalog.arizona.edu/policies

Campus Health
http://www.health.arizona.edu/

Campus Health provides quality medical and mental health care services through virtual and in-person care.

- Phone: 520-621-9202

**Counseling and Psych Services (CAPS)**
https://health.arizona.edu/counseling-psych-services

CAPS provides mental health care, including short-term counseling services.

- Phone: 520-621-3334

**The Dean of Students Office's Student Assistance Program**
http://deanofstudents.arizona.edu/student-assistance/students/student-assistance

Student Assistance helps students manage crises, life traumas, and other barriers that impede success. The staff addresses the needs of students who experience issues related to social adjustment, academic challenges, psychological health, physical health, victimization, and relationship issues, through a variety of interventions, referrals, and follow up services.

- Email: DOS-deanofstudents@email.arizona.edu
- Phone: 520-621-7057

**Survivor Advocacy Program**
https://survivoradvocacy.arizona.edu/

The Survivor Advocacy Program provides confidential support and advocacy services to student survivors of sexual and gender-based violence. The Program can also advise students about relevant non-UA resources available within the local community for support.

- Email: survivoradvocacy@email.arizona.edu
- Phone: 520-621-5767

**Support Outreach Success (SOS)**
https://sos.arizona.edu/about

Students who ask for help are more successful. If you have questions, concerns or challenges and are unsure about where to go for answers or support you can ask SOS. The SOS staff will answer questions, find you resources or connect you with the correct people. Whether you're brand-new to campus or have been around for a while, just reach out to SOS for support.

- Chat: sos.arizona.edu
- Email: sos@arizona.edu
- Text: SOS to 70542
- Call: 520-621-2327

**Confidentiality of Student Records**
http://www.registrar.arizona.edu/ferpa

**Safety on Campus and in the Classroom**
For a list of emergency procedures for all types of incidents, please visit the website of the Critical Incident Response Team (CIRT): https://cirt.arizona.edu/case-emergency/overview
Also watch the video available at
https://arizona.sabacloud.com/Saba/Web_spf/NA7P1PRD161/common/learningeventdetail/crtfy000000000003560

**Subject to change statement**
The information contained in this syllabus, other than the grade and absence policies, may be subject to change with reasonable advance notice, as deemed appropriate by the instructor.